

Curriculum Vitae

Tom Wiesing | tom@tkw01536.de | <https://github.com/tkw1536/> | Erlangen, Germany

Stack Go | Python | Django | TypeScript | React | Vite | HTML | CSS | SQL | Postgres | SQLite | SPARQL

Tooling Docker | Docker Compose | Git | GitHub Actions | Linux Sysadmin | Ansible | Traefik | Nginx

Personal Projects

Alumni Portal (Django + TypeScript) - Alumni Management System
volunteer work

jay (Django + JavaScript) - Student/Alumni Voting System
still in use after 10+ years

ggman (Go) - Repository Manager

Poke The Game (Node + JavaScript)
Winning Project for JacobsHack 2014!

lots of smaller projects - most based on React + TypeScript

Professional Experience

Software Engineer - Competence Center for Research Data and Information

09/2021 - present | Friedrich-Alexander-Universität Erlangen, Germany

- WissKI (Drupal + PHP + RDF + SPARQL) | TIPSY (React + Typescript + Vite) | lontod (Python + Sqlite + RDF) | quickpid (Go) | my dissertation & projects
- Developed and maintained research data management systems used across academic projects.
- Co-led interdisciplinary teams to design, build & integrate software solutions into research workflows.
- Led deployment pipelines to configure, document & maintain infrastructure using Docker, Ansible & Git.

Research Software Engineer - KWARC Group

09/2017 - 08/2021 | Friedrich-Alexander-Universität Erlangen-Nürnberg, Germany

previously **07/2015 - 05/2017** | Jacobs University Bremen

- MathHub (Next + TypeScript) | localmh (Python + Docker)
- Built knowledge representation tools and scientific knowledge services.
- Implemented backend systems for semantic markup and interactive publishing.
- Maintained & developed tools to support formal and informal mathematics research.

Research Software Engineer (Exchange Visitor) - Applied and Computational Mathematics

Division - Mathematical Software Group

06/2021 - 08/2021 | National Institute of Standards and Technology (NIST), Gaithersburg, MD, USA

also **11/2019 - 01/2020, 06/2018 - 08/2018, 06/2017 - 08/2017**

- Support for various LaTeX packages in LaTeXML (Perl + LaTeX + XSLT)
- Setup Docker-based CI & Testing Infrastructure for LaTeXML.

Education

Doctor Of Engineering

2017 - 2025 | Friedrich-Alexander-Universität Erlangen-Nürnberg

WissKI Distillery (Go + Docker + Typescript + PHP) | MathDataHub (Django + React + Postgres)

Dissertation: *Towards A Commons Of Object-Centered Research Data* (doi: [10.25593/open-fau-1825](https://doi.org/10.25593/open-fau-1825))

Master Of Science in Data Engineering

2015 - 2017 | Jacobs University Bremen (now Constructor University Bremen)

Bachelor of Science in Applied & Computational Mathematics

2012 - 2015 | Jacobs University Bremen (now Constructor University Bremen)